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Optimizing Seismic Monitoring Strategies in Romania's Western Black Sea with 4D Full Waveform Inversion

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Production monitoring through time-lapse seismic has been a common practice for decades in the oil and gas industry and is used as a key technique to protect investments, by extending field life in identifying by-passed reserves, and facilities via early detection of water break-through. Parallel to its increased utilization, there has been continuous evolution in seismic acquisition and processing techniques, making it progressively possible to obtain reservoir monitoring at smaller scales and in assets with increased geological complexity.

A large deep-water gas field in the Romanian Black Sea is currently undergoing development drilling and facilities installation. The subsea installations are particularly challenging due to very rugose and unstable seabed topography and very low water temperatures, thus triggering large investment in construction and facilities. To ensure the field's economic viability, it is critical to achieve a sustained and high production plateau. The seismic interpretation suggests that the reservoir is compartmentalized making it essential to identify untapped compartments during production. Timely identification of these compartments using time-lapse seismic data would enable infill drilling to target untapped gas reserves and maintain production plateau stability.

Planning and executing a cost optimal time-lapse seismic survey, which can accurately deliver information at the appropriate time to guide infill drilling using seismic history matching workflows is essential. It requires detailed assessment of the field specific expected time-lapse signal signature and magnitude, the associated noise levels and the seismic acquisition configuration to evaluate the signal acquisition over the noise.

Intro

Production monitoring through time-lapse seismic has been a common practice for decades in the oil and gas industry and is used as a key technique to protect investments, by extending field life in identifying by-passed reserves, and facilities via early detection of water break-through. Parallel to its increased utilization, there has been continuous evolution in seismic acquisition and processing techniques, making

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This paper describes the workflow developed to optimize surface seismic monitoring for the field. The workflow incorporates classical seismic modelling techniques with advanced modelling and analysis to assess whether new technologies such as 4D FWI can enhance the detectability of 4D seismic signal where it is of a similar magnitude to the estimated noise level. By deriving these insights, the workflow ensures that acquisition timing and efficiency are optimized, thereby increasing the likelihood of successful reservoir monitoring and informed decision-making.

Initial framework

The field undergoing development is located in a concession area in Western Black Sea, Offshore Romania (Figure 1). The sedimentary units mainly consist of Miocene turbiditic sequences including channel-levee and lobe deposits. Mass transport complexes (MTCs) in the overburden pose challenges for conventional seismic imaging. While there are no carbonates in the overburden, the presence of shallow gas accumulation causes significant absorption in addition to overburden transmission losses and scattering. The seabed is rugose due to submarine landslides and deeply incised canyons in the area.



Figure 1—Location of the study area

The production is expected to be driven primarily through depletion rather than water drive. A strong waterfront is not expected, nevertheless potential pathways including turbiditic channels connect the gas reservoir with downdip brine-filled sandy units. Therefore, pressure changes are expected to be the primary change to be observed with time-lapse seismic: an uneven pressure depletion is considered to be an indication of potential left-behind portions of reservoir which need to be detected early to plan for infill drilling. In addition, early water breakthrough represents the main risk to be detected and mitigated.

The subsurface is imaged by a towed streamer seismic survey acquired in 2012, with subsequent reprocessing efforts. This survey was acquired with 4D in mind by shooting with overlapping streamers. The $12 \times 6000\text{m}$ streamer configuration used was acquired on an 8-streamer pre-plot. Five exploration wells penetrated the reservoir with complete log suite (gamma ray, neutron, bulk density, caliper, P-sonic and S-sonic).

To address these challenges, the following questions primarily need to be quantitatively answered:

- Is the reservoir response to predicted production amenable to future 4D monitoring?
- If yes, which realistically recordable survey designs would be suitable for use as a future 4D monitor?
- Is the legacy survey suitable as a future time-lapse survey baseline?
- Can a cost-effective candidate survey be designed to meet these challenges?
- Can a future acquisition be de-risked based on the known challenges of the subsurface?

To answer these questions, a workflow touching 4D Rock Physics modelling, Geometry Design and 4D detectability assessment was designed and executed (Figure 2).

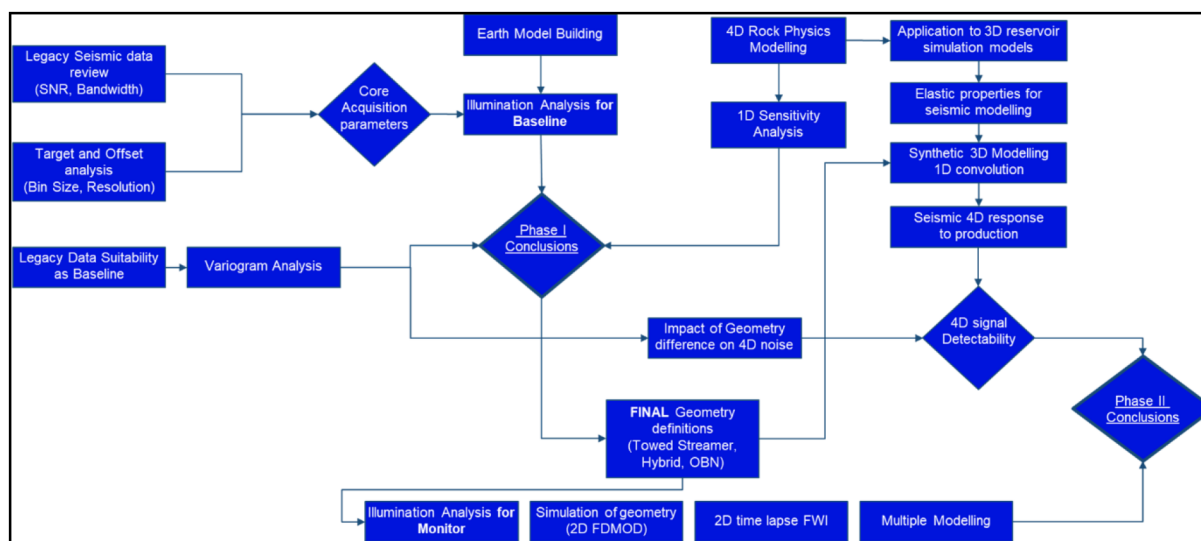


Figure 2—Schematic of the workflow implemented in this study

4D rock physics and simulation-to-seismic modelling

The rock physics modelling predicts elastic and acoustic properties from clay volume, porosity, pressure and varying saturations of brine and gas. A stress dependent field specific rock physics model was calibrated using the available well dataset. After this the simulation to seismic modelling workflow was applied. The calibrated rock physics model was used to forward model the acoustic and elastic properties in the reservoir intervals in 3D from the available static and dynamic models. Combined with the elastic properties from

pre-stack seismic inversion of the baseline seismic data for the non-reservoir intervals full 3D volumes of synthetic seismic were modelled for key production years.

Time lapse interpretation of the 3D modelled results enable changes in the acoustic and elastic properties to be evaluated as a function of the percentage change in pressure, considering different pressure and saturation scenarios: 1) pre-production, 2) after initial production period and 3) at the later stage of production plateau.

The modelling results indicate the following changes:

- Pressure depletion after initial production causes 2.5%-4.5% of Acoustic Impedance difference and P-velocity change of +50-80m/s
- Pressure depletion at later production stage causes 4.5%-7.5% of Acoustic Impedance difference and P-velocity change of +80-150m/s
- Water saturation changes cause Acoustic Impedance difference from 0-50% and P-velocity difference +0-550m/s depending on the magnitude of the sweep.

It can be observed that the reservoir has a detectable 4D acoustic impedance change to fluid saturation changes from baseline conditions. However, pressure sensitivity is only moderate to weak in case of pressure depletion when not combined with a saturation change. The sole pressure depletion signal results in a maximum Acoustic Impedance change of 7% from the pre-production baseline (Figure 3). This low percentage value is unlikely to be detectable above 4D noise levels for 4D towed streamer acquisition where global data indicates Normalized Root Mean Square (NRMS) ~15%.

Percent Change	Baseline to Initial Production Stage	Baseline to Advanced Production stage
Delta Saturation	0	0
Delta Reservoir Pressure	-7	-39
Max. Delta Acoustic Impedance	4.7	7.0
Max. Delta P-velocity	5.3	8.2
Max. Delta Density	-0.5	-2.0
Max. Delta Vp/Vs	-1.0	-1.5
Max. Delta S-velocity	6.4	9.8

Percentage change = (Monitor - Baseline / Baseline) * 100
 where positive value is an increase, and negative value is a decrease
 between the baseline and monitor

Figure 3—Percentages changes of Reservoir and Elastic properties due to production pressure depletion only, as derived with the Rock Physics Model

This early indication clearly poses challenges to the production monitoring aspirations and requires a careful assessment through appropriate seismic modelling to identify if a particular cost-effective configuration of 4D seismic acquisition and processing could retrieve enough pressure depletion signal while mitigating the noise.

4D survey design and acquisition strategy

To identify candidate monitor survey designs for time lapse scenarios and gather necessary information, an initial review of existing surveys and related surface and subsurface data was conducted.

The assessment of the quality and suitability of the existing survey to act as a baseline in general concluded that it offers adequate repeatability in 4D context and is therefore appropriate as a baseline.

More in detail, the ray tracing exercise simulating compressional reflection ray paths indeed confirmed that a maximum offset of 6 km, as present in the existing survey, is adequate for imaging at reservoir depths and provides up to 60-degree incidence angles to enable AVO analyses.

Yet long offsets are crucial for recording refracted and diving waves which are key to full-waveform inversion (FWI), FWI captures the low-frequency and low-wavenumber components of seismic velocities that cannot be obtained from reflection-only data. This, in turn, is key for characterising deeper geological structures with correct kinematics. (Figure 4). Various publications have shown improvements to the seismic data quality and the vertical resolution can be realised by improving the accuracy of the migration velocity (Chapelle et al., 2023), and an accurate, high-resolution velocity model is deemed key for time-lapse imaging and velocity model building where expected elastic changes from production may be below or close to the noise floor of the acquired seismic data (Dutta et al., 2023).

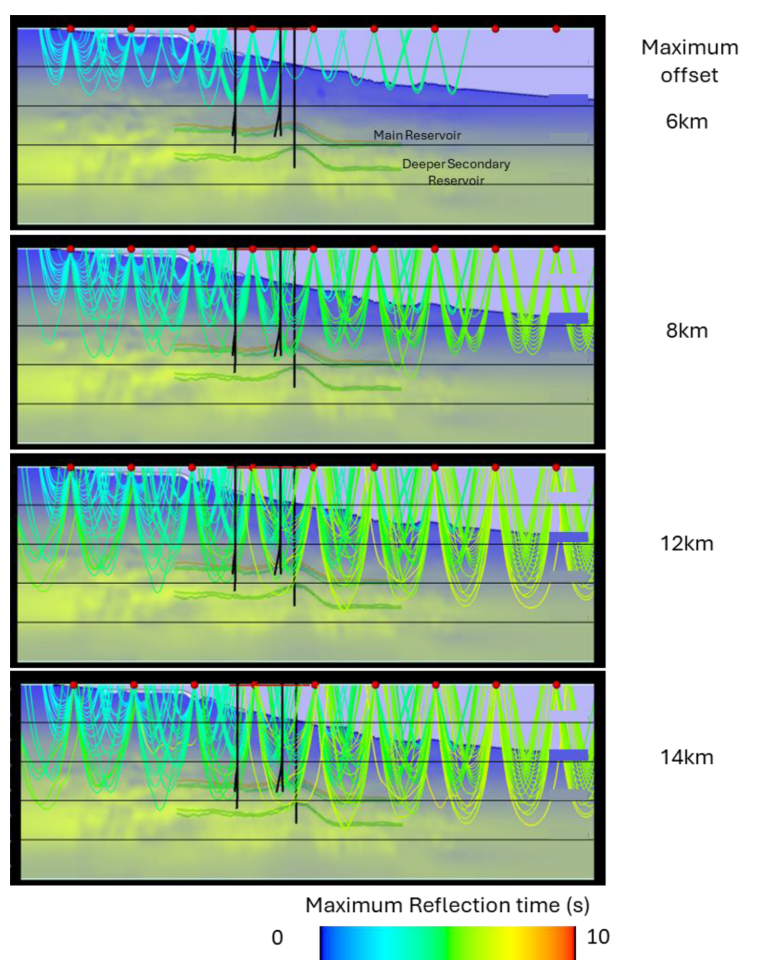


Figure 4—illustrates Longer offsets (than the legacy) would benefit the FWI velocity update the legacy maximum offset is 6 km and offers penetration to around 1.5-2km depth. (good match with the presence of details in the legacy velocity); When increasing the maximum offset the penetration depths increases (around 2.5-3km dept at 8km) and a plateau can be observed from maximum offset of 12-14km

For this purpose and following a survey design exercise, practical monitor acquisition designs options were developed alongside diving wave ray tracing and a synthetic diving wave FWI simulation exercise based on input acoustic synthetic shot gathers derived from the legacy velocity model, simulation-to-seismic exercise and simulated shallow reflectivity from post-stack AVO inversion, which altogether led to the following options (Figure 5)

- Repeating the existing towed streamer survey with 6 km maximum offset
- Repeating the existing towed streamer survey with 6 km maximum offset, with two long streamers up to 10 km
- Repeating the existing towed streamer survey with 6 km maximum offset combining a hybrid survey with a grid of ocean bottom nodes (OBN) at 600m, 900m or 1200m
- Repeating the existing towed streamer survey with 6 km maximum offset combining a hybrid survey with a grid of ocean bottom nodes (OBN) at 600m, 900m or 1200m and with two long streamers up to 10 km
- Transition to a dense OBN baseline and monitor survey configuration

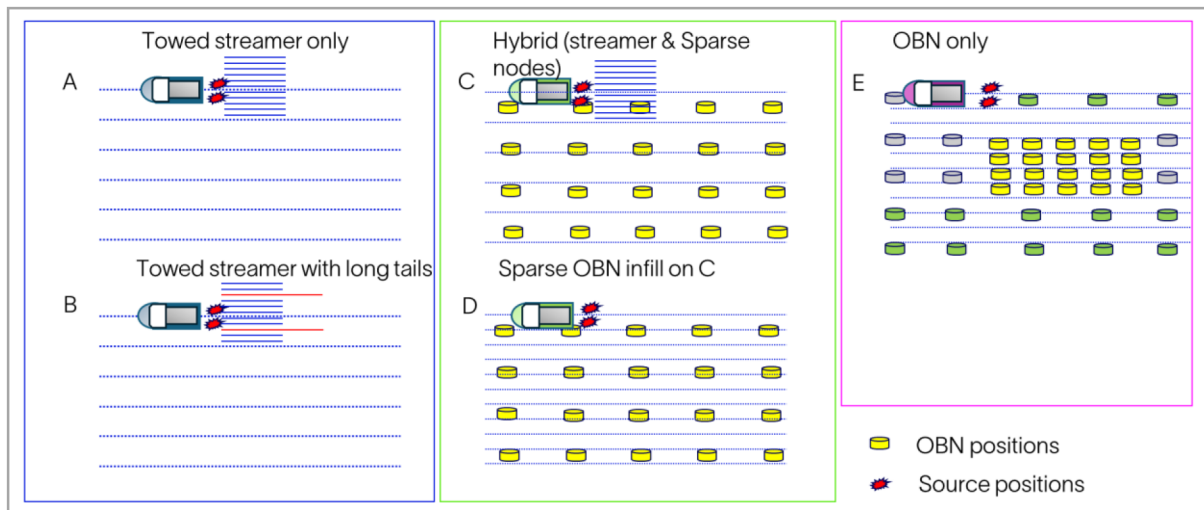


Figure 5—Acquisition geometries considered for benchmark: A = towed streamers all 6 km; B= towed streamers 6 km plus two long tails 10 km; C = towed streamers 6 km plus sparse OBN; D = infill of the OBN in the core area, to be intended as infill if C survey; E = OBN only survey, to be intended as monitor survey after first hybrid acquisition (C)

There are three main reasons mentioned by operators for transitioning to OBN as part of a 4D seismic program:

1. Magnitude of 4D signal expected: OBN allows better repeatability allowing to detect weaker 4D signals (Davies et al., 2020).
2. Type of 4D signal: OBN can record PP and PS. P-waves (compressional waves) amplitude and velocity change with different types of fluids (e.g., gas, oil, water) and their saturation levels. On the other hand, S-waves (shear waves) are primarily affected by the rock matrix and its mechanical properties. Comparing PP data (involving only P-waves) and PS data (involving both P-waves and S-waves) allow discrimination between pressure and saturation effect: PS data, which involves both P-waves and S-waves (Tura et al., 2024).
3. Increased SIMOPS flexibility (Simultaneous Operation encompasses operations that occur at the same time than surveying and would interfere with seismic acquisition), (Burnett et al, 2025). Towed streamer surveys are impacted by exclusion zones around surface infrastructure to avoid collisions by

the streamers and results in areas where data cannot be acquired. Unlike towed streamer vessels, OBN configurations can be deployed close or below infrastructures and the source vessel benefits of more flexibility in terms how close it can bring the source to the obstructions than a towed streamer vessel.

One of the main benefits of 4D OBN survey when compared 4D towed streamer surveys come from the fact that they achieve a higher repeatability of both the source and the receiver. OBN survey shooting patterns could enable a source grid of $25\text{m} \times 25\text{m}$, ensuring an optimum bin size of 12.5m and the dS (delta source) position will never be larger than 12.5 for reduced NRMS noise.

However, as production is driven primarily by depletion and with no injector wells planned at the time of this study, the need to differentiate between pressure and saturation changes is less critical and the requirement for PS data less crucial.

On the infrastructure aspect, the field also creates a special case given the planned subsea template and being production facilities located farther away in shallow water, therefore no obstructions will be present over the expected survey area.

Because of those facts and the presence of the legacy survey already acquired with 4D in mind, our conclusion was to continue a monitoring strategy using towed streamer acquisition until infrastructure plans change.

The remaining monitor survey options were benchmarked against expected costs and a quantitative scorecard to measure improvement in velocity modelling and illumination for each geometry for five intervals between seabed to reservoir (Figure 6), with the scores relative to the best performing geometry for each test. The long tails towed streamer acquisition has proved as the most cost-effective manner to repeat the existing survey whilst also collecting long offsets for diving waves.

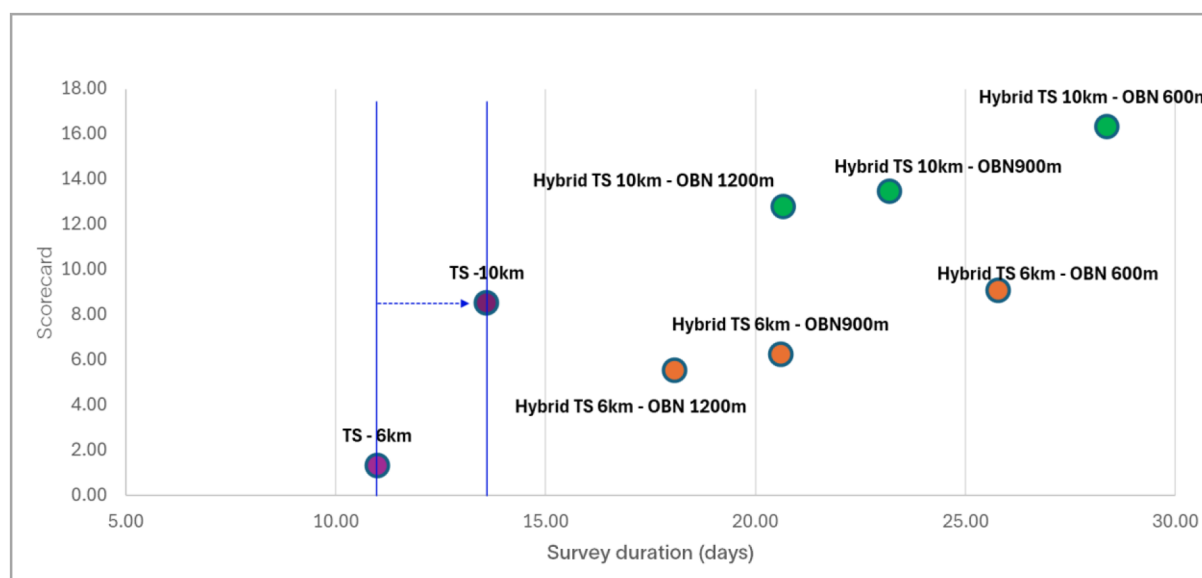


Figure 6—Benchmark chart for the acquisition geometries considered. The scorecard in Y-axis represents a combined value which accounts for illumination capability and velocity resolution at five different depths (from near seabed to the deeper secondary reservoir). Each single parameter/depth is given a normalized score between 0 (poor) and 1 (optimal); for the hybrid surveys the actual scorecard is the sum of the two components

Referencing the questions defined at the beginning of this article, these experiments have defined realistically recordable survey designs, suitable for use as a future 4D monitor, based on the assessment of the legacy survey being appropriate as a future time-lapse survey baseline, with alternate survey configurations quantitatively evaluated to define cost-effective candidate surveys.

The 4D rock physics modelling and simulation-to-seismic modelling suggested the reservoir response was of sufficient magnitude to be amenable to future 4D monitoring, and appropriate surveys have been

designed. These two items are combined to de-risk future acquisition based on the known challenges of the subsurface, defined by the reservoir model, for detectability evaluation in terms of image and data domain time-lapse seismic methods in the presence of noise.

4D detectability in the image domain in the presence of noise

The rock physics modelling highlighted that the pressure depletion signal is relatively weak and may lie close or below the noise level. Amplitudes related to the 4D signal and the 4D noise can be quantified using the Normalized Root Mean Square (NRMS).

Using the rock physics model, a simulation-to-seismic analysis enabled convolutional stack volumes to be generated for specific time scenarios mentioned above (pre-production, after initial production period and later stages of production plateau) and 4D amplitude differences were calculated in between these steps. It is now useful to recall the signal differences in terms of NRMS:

- Pressure depletion after initial production is equivalent to NRMS from 3% to 10%
- Pressure depletion at later production stage is equivalent to NRMS from 5% to 15%
- Water saturation change is equivalent to NRMS from 0% to 75% depending on the magnitude of the sweep.

The legacy survey assessment and the survey design have estimated the 4D noise to be 25% NRMS for the scenario where the baseline is repeated by a monitor with the same geometry. This is however considered a pessimistic scenario, considering literature suggesting on global average a 4D noise level of 15% NRMS for towed streamer acquisition, of 12% NRMS for Ocean Bottom Nodes and 5% for Permanent Reservoir Monitoring (Macbeth et al., 2024).

To quantitatively assess the 4D detectability in presence of noise, a noise model was scaled to achieve a RMS amplitude equal to 5%, 10%, 15%, 20%, 25%, 30% of the average RMS value of the survey and added to the 4D difference stack volume to represent the background 4D noise (see Dhelie et al., 2015).

The results were generated on maps and on volumes and can be summarized as follows:

- Pressure changes would not be detectable based on amplitudes differences for the Towed Streamer monitoring strategy due to the low NRMS of this signal (<15%) at least until advanced production stages, although recently published case studies have demonstrated better outcomes in similar conditions (Borgmeyer et al. 2024).
- Gas water contact movement will be detectable based on amplitude differences.

Leveraging full waveform inversion: 4D detectability in the data domain in the presence of noise

The detectability assessment conducted on convolution stacks has concluded that a monitoring strategy based time-lapse amplitude variation will likely not provide the steering needed to execute infill drilling. However, it should be noted that detectability on 1D convolution stacks relies on looking at amplitude differences (driven by the change of AI, density and Vp/Vs ratio between the different years) but the 1D convolution stack have no timing differences in the positions of the events (perfect alignments). Being the production occurring in deep water turbidite bodies, characterized by softer lithologies, it is expected that compaction will occur as consequence of pressure depletion.

We therefore decided to explore an alternative detection approach based on 4D Full Waveform Inversion (4D FWI). to directly find the 4D difference in the model domain. For example, Dutta et. al. (2023) introduced a 4D FWI approach that requires minimal data preprocessing and minimizes acquisition-related

differences by solving for both the baseline and the monitor models in a joint inversion (Halliday et al., 2024).

Conventionally, FWI has been used to invert for velocity models and thus to improve imaging. In the last decade or so, however, vastly increased computational power enabled running FWI to very high frequencies, making it an excellent strategy for capturing time-lapse velocity changes as well (Dutta et al., 2023). One of the key characteristics of 4D FWI is its ability to detect production-induced velocity changes directly, potentially eliminating the need for time-shift or depth-shift alignment techniques typically used in 4D analysis. In conventional workflows, baseline and monitor seismic images are aligned using calculated shifts before any time-lapse differences are assessed. However, with 4D FWI, each data set is migrated with its corresponding velocity model, making separate alignment steps unnecessary. Such direct capture of 4D velocity changes has the potential to significantly enhance the quality and reliability of 4D seismic data as well as their interpretation (Albertz et al., 2025).

To evaluate and predict the effectiveness of 4D FWI to detect predicted production changes, the 3D static and dynamic model of the field was used to drive a synthetic experiment. With assumption of an accurate starting elastic earth model, multiple compressional velocity fields were created to represent subsurface changes over time for baseline and monitor timesteps, with velocity and density changes over time taken directly from the 4D rock physics and simulation to seismic modelling. These high-resolution models were used to derive acoustic finite difference forward modelled shots up to 40Hz for each candidate geometry and each timestep within a 2D experiment.

For the baseline timestep two geometries were produced for each synthetic expressed in terms of dSdR which indicates the sum of the differences in source position (delta-source, dS) and receiver position (delta-receiver, dR):

- dSdR = 0 uses the same location for the baseline and the monitor geometry, illustrating a perfect repeatability
- dSdR = 25m the second with bulk shifted by 25m between baseline and monitor

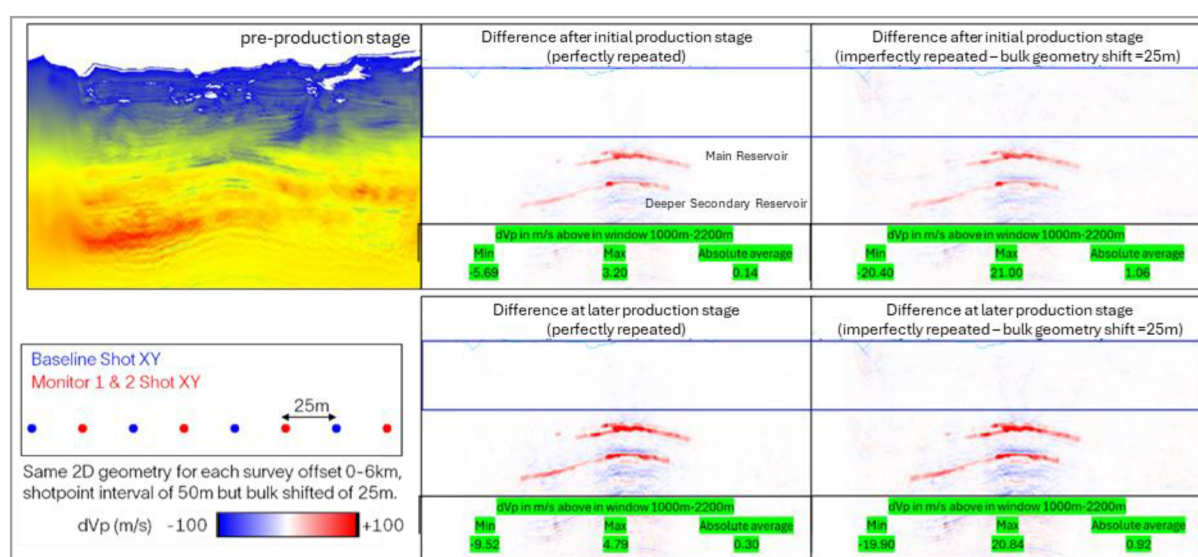


Figure 7—4D FWI/FDR tests comparing the FWI response of a perfectly repeated geometry to an imperfectly repeated geometry with a dSdR of 25m

Finally, the FWI workflow was initiated using a smoothed version of the baseline velocity model (pre-production stage). The inversion on these synthetic data pairs was run in parallel for the baseline and two monitor stages.

FWI was progressively evaluated in several bands from 5Hz, 8Hz, 12Hz, 17Hz, and 23Hz using combined refraction and reflection wavefields. While the experiment could have continued with higher frequencies, it was stopped at 23Hz as this allows several conclusions to be drawn on the suitability of the acquisition geometry and the magnitude of the velocity change that 4D FWI can detect.

The main conclusion drawn from observing the results is that 4D FWI can detect the variations in areas where the change of velocity is as low as 50m/s between baseline and monitor.

For the case with artificially introduced geometry difference of $dSdR=25m$ it was observed that the delineation of the 4D signal on P-velocity was not affected, although it caused a background dVp not related to the outside the reservoir (increases background dVp from 0.14m/s to 1.06m/s after initial production and from 0.3m/s to 0.92m/s at later stage of the production (not shown).

FWI Derived Reflectivity (FDR)

The velocity difference (dVp) obtained from parallel 4D Full Waveform Inversion (FWI) can be transformed into FWI-Derived Reflectivity (FDR). This approach has gained traction in recent years as FDR provides an alternative output that holds significant potential for seismic interpretation. One of the key advantages of the FDR methodology is that it bypasses the need for a full seismic processing workflow, enabling direct computation from the shot domain to a property volume in depth. In this process, Vp is used as a proxy for acoustic impedance, recognizing that a "true" reflectivity would ideally be derived from the derivative of impedance, for this comparison we are computing the spatial derivative of dVp .

In this study, the pseudo 4D FDR was evaluated by comparing it to the 4D stack difference, which was calculated by subtracting the 1D convolutional stacks generated for different years. To ensure a robust comparison, the 4D stack difference was assessed both with and without the addition of random noise.

The results determine the 4D detectability levels achievable through 4D FWI-derived outputs compared to those enabled by traditional image domain interpretations. This comparative approach highlights the potential of 4D FDR to provide improved resolution and interpretability while streamlining the workflow by eliminating the need for conventional seismic processing steps. These findings emphasize the value of FDR as a promising tool for advancing seismic interpretation in time-lapse studies.

While the 4D FWI is not a perfect reconstruction and presents a background "ringing", we observe a clear velocity increase as expected due to the modelled caused by either the pressure depletion and the water replacing the gas. More in detail, FDR appears to have the ability to detect change related to pressure depletion equivalent to $dVp = 50-150m/s$ (Figure 8). A visual comparison of 4D FDR with the 4D stack difference with noise suggests an equivalent detectability with what is achieved on amplitude with 4D noise of 5% or 10% suggesting that 4D FWI would provide an improvement over detecting using 4D stacks (remember: global data for a 4D towed streamer acquisition is 4D noise = 15% and our pessimistic case of 4D noise of 25% from the CMP variograms). Published examples on real data support the conclusions that 4D FWI provides an improved signal-to-noise ratio and improve 4D detectability (Dutta et al., 2023). It should be noted that, from a resolution point of view, the 4D FDR product appears to have less resolution than the 4D stack difference noise free but this would likely to be solved by carrying toward higher frequency bands 4D FWI at higher frequencies than 23 Hz.

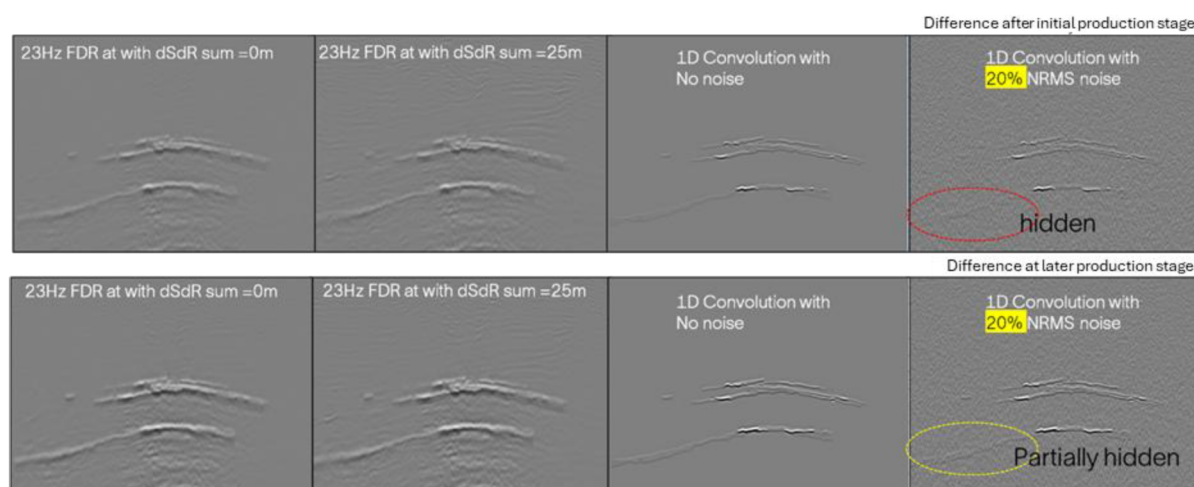


Figure 8—4D FWI/FDR tests: comparison of FWI FDR with 1D convolution modelling after initial production stage and after later production stage with and without the presence of noise

Acquisition strategy for 4D FWI

One key conclusion of our modelling study is that detectability using 4D FWI is achieved using a maximum offset of 6 km although it does not collect refraction reaching the main reservoir. We indeed showed that 4D FWI would enable detection of the change in the reservoir for the magnitude predicted by the 4D rocks physics model with the legacy configuration (6 km) but the availability of an accurate starting velocity model is a paramount component of this achievement.

Our conclusion is therefore that the presence of long offsets is not a requirement for 4D detection but only for the 3D velocity model building and that those offsets absent from the legacy survey are not critical for 4D FWI.

Conclusions

Using a calibrated rock physics and simulation-to-seismic exercise, the study has characterized two types of 4D signals in the reservoir:

1. A pressure depletion signal
2. A saturation change signal (water replacing the gas)

This has been combined with a thorough seismic survey analysis and design exercise to summarize and conclude the following surface seismic acquisition strategies for monitoring and characterization objectives.

Two strategies were considered to detect the 4D signal during the survey design evaluation:

1. A conventional 4D amplitude detection through seismic data processing and imaging.
2. A 4D Full Waveform Inversion (FWI) strategy, including FWI derived reflectivity (FDR).

We conclude that the pressure depletion signal will not be confidently detectable based on 4D amplitude changes with a Towed Streamer on Towed Streamer (TS-on-TS) monitoring strategy using image-based 4D seismic processing. This is due to the relatively small magnitude of the pressure depletion signal on the seismic wavefield being associated with an Acoustic Impedance (AI) change of 2.5%–7.5%, corresponding to a Normalized Root Mean Square (NRMS) value of 3%–15%. However, this signal lies below the expected noise floor for a TS-on-TS strategy, where 4D noise (as defined by the NRMS metric) is typically 15% or higher. A recent literature example for detection of pressure changes due to depletion in clastic reservoirs however casts a more favorable perspective: the authors were able to infer pressure depletion due to a 5.5% change in AI over a similar pressure decrease and an observed NRMS of 14% (Borgmeyer et al. 2024).

Nevertheless, we consider the detection of this signal would be unlikely for our case, as the 4D noise for this field is estimated to be 25% NRMS, or at least dependent on a seismic acquisition program that maximizes source and receiver repeatability, in favorable conditions.

We have analyzed 4D FWI as an alternative time-lapse seismic data processing strategy. Our modeling suggests a higher detectability level can be achieved than with conventional image-based seismic processing of data. The FWI results were modelled with noise-free data but suggest the velocity changes associated with the pressure depletion could be detected. This makes 4D FWI a recommended monitoring strategy.

With regards to the acquisition strategy, we conclude that the legacy survey is adequate for imaging the time-lapse signal of the reservoir. It is feasible to repeat this configuration for future time-lapse surveys, with some enhancements. Specifically, the deployment of longer streamers, at least 10 km in length, is critical for supporting velocity model building. This improvement is particularly important for a successful 4D FWI strategy, to capture diving waves necessary for FWI velocity model update.

A hybrid survey including both Towed Streamers and sparse Ocean Bottom Nodes does not appear to add significant uplift. The OBN would indeed offer an attractive aid in building the velocity model however it is unlikely to generate an imaging product with better resolution than the imaging product from the towed streamer dataset. Furthermore, the increased costs and longer turnaround times associated with the hybrid solution are deemed unjustified in context of this field development. Therefore, a pure towed streamer 4D acquisition strategy, optimized for repeatability and enhanced streamer length, remains the most cost-effective and practical approach for seismic reservoir monitoring during production for this field.

4D monitoring for saturation change is not discussed in detail in this paper as the changes are significantly above any expected NRMS and likely detectable in most 4D seismic scenarios.

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